Multivariate Lab-6 **Date: 10-03-2026**

Perspiration from a sample of 20 healthy females was analyzed. Three components, X_1 = Sweet rate, X_2= Sodium content and X_3=Potassium content, were measured and the results, which we call the sweat data are given in the below table.

Sweat Data			
Individual	X_1 (Sweat rate)	X_2 (Sodium)	X_3 (Potassium)
1	3.7	48.5	9.3
2	5.7	65.1	8.0
3	3.8	47.2	10.9
4	3.2	53.2	12.0
5	3.1	55.5	9.7
6	4.6	36.1	7.9
7	2.4	24.8	14.0
8	7.2	33.1	7.6
9	6.7	47.4	8.5
10	5.4	54.1	11.3
11	3.9	36.9	12.7
12	4.5	58.8	12.3
13	3.5	27.8	9.8
14	4.5	40.2	8.4
15	1.5	13.5	10.1
16	8.5	56.4	7.1
17	4.5	71.6	8.2
18	6.5	52.8	10.9
19	4.1	44.1	11.2
20	5.5	40.9	9.4

Population mean vector for Sweat rate, Sodium content and Potassium content is given as $\mu_0' = [4, 50, 10]$. Test the hypothesis $H_0 : \mu' = \mu_0'$ against $H_1 : \mu' \neq \mu_0'$ at given level of significance $\alpha = 0.05$ [$F_{(3,17)}(0.05) = 3.197$].

Step 1 :- Hypothesis Satements :

Null Hypothesis :- There is no significance difference between Sweat rate, Sodium Content and Potassium content .

$$H_O : \grave{\mu} = \grave{\mu}_o$$

Alternate Hypothesis :- There is significance difference between Sweat rate, Sodium Content and Potassium content .

$$H_1 : \grave{\mu} \neq \grave{\mu}_o$$

Step 2 :- Data Analysis :

```
library(readxl)
problem_6 <- read_excel("D:/MULTIVARIATE WORK/problem 6.xlsx")
x1<-problem_6$`X1 (Sweat rate)`
x2<-problem_6$`X2 (Sodium)`
x3<-problem_6$`X3 (Potassium)`
n<-20

data_matrix<-matrix(data=c(x1,x2,x3),ncol = 3)

cov_matrix<-cov(data_matrix)

mean_vector<-colMeans(data_matrix)

inverse_s<-solve(cov_matrix)

m_note<-matrix(data=c(4,50,10),nrow = 3)
m_note

x_bar_m_note_transpose<-t(mean_vector-m_note)

x_bar_m_note<-(mean_vector-m_note)

t_sq<-20%*%x_bar_m_note_transpose%*%inverse_s%*%x_bar_m_note
t_sq

##          [,1]
## [1,] 9.738773

t_cal<-(17/(19*3))*t_sq
t_cal

##          [,1]
## [1,] 2.904546
```

Step 3 :- Conclusion :

At the 5% level of significance :

T-calculated = 2.904 < 3.197 .

So, We failed to reject the null Hypothesis ,

The sample mean vector of sweat rate, sodium content and potassium content **does not differ significantly** from the population mean vector.